What a reward model can show depends on who stopped the episode: failure judges and monitors on a low-cost arm

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Abstract

Reward models that score robot rollouts are also used as failure detectors. We compared two zero-shot reward models with plain episode length. On ArmnetBench, a public SO-101 benchmark where runs end at a time limit, length separates failure from success at AUROC 0.942 within policy. Failed runs are slow rather than cut off. Length observed only up to half of the longest episode still gives 0.851. Both zero-shot reward models score below length, the better one at 0.821. The released fine-tune, scored on its own training episodes, reaches 0.994. On our own rollouts an operator stopped each run by judgment. There length falls to 0.56, and neither reward model beats the identity of the policy. A judge that sees the finished episode passes its threshold, at an operating point chosen in-sample. No mid-episode monitor reaches its bar. The baseline a detector has to beat depends on the dataset's stopping rule.

1 Introduction

Papers that evaluate a reward model as a failure detector normally compare it with other learned detectors. When a benchmark stops failed runs at a time limit, the length of an episode also encodes its outcome. The identity of the policy that ran can carry it too, and we call this the policy-identity null (mainly with models that give out-of-distribution results). When the operator stops each run by his judgment, both signals are weaker. We call the way a dataset ends its episodes its stopping rule. None of the reward-model papers in our library reports the stopping rule or a length baseline.

We started with a narrower claim, that elapsed time is the reward model to beat. Our own rollouts did not support it. Length is near chance on them. A robot-trained judge passed its pre-registered threshold on them. What separates the two results is the stopping rule, so it became the subject of the paper.

We measured length on ArmnetBench [12] and on our own rollouts. The length analyses were frozen before any number existed. We compared length with two zero-shot reward models, Robometer [5] and TOPReward [1]. On our 88 operator-stopped rollouts we also compared it with every judge and monitor we had run.

2 Data and labels

ArmnetBench v0.1 [12] has 8 tasks and 7 policies on the SO-101. We use its 2,047 policy rollouts that have a success or failure class (515 successes, 1,532 failures) and leave out the teleoperated episodes. One task, pushing a Displayport cable (270 rollouts), has rollouts of a single class, so it has no per-task AUROC. Our own data are 88 labelled bowls-task episodes on one SO-101 arm with a Jetson Orin NX. Every monitor gate used the same 50-episode subset (42 successes, 8 failures). The author was the operator, stopped each run and labelled it at rest, and was not blind to the policy.

3 The length baseline

On ArmnetBench, episode length is a strong failure detector within each policy (Table 1, Fig. 1). Pooled over policies it reaches 0.938. Within each policy it ranges from 0.91 to 0.96. Pair-weighted, the within-policy value is 0.942, with a CI of 0.927–0.956.

The within-policy value pair-weights the per-policy AUROCs (`repro/length-null/scripts/length_analysis.py`):

```
def stratified(df, score, by="policy_type"):
    """Pair-weighted mean of per-stratum AUROCs (weights successes
      x failures)."""
    num = den = 0.0
    for _, g in df.groupby(by):
        a = auroc(g.y, g[score]); w = g.y.sum() * (~g.y).sum()
        if a is not None:
            num += a * w; den += w
    return num / den if den else None
```

We ran Robometer-4B [5] zero-shot, as its authors ship it. It scores 0.818 pooled and 0.821 within policy (CI

*Code, result files and the checks that regenerate every number: `github.com/ravediamond/paper-reward-models-stopping-rule`. The frozen protocols are copied verbatim under `repro/`; the bar commits named in the paper belong to the authors' research repository `ravediamond/road-to-robotics`.

signal	pooled	within policy
episode length	0.938 (0.925–0.951)	0.942 (0.927–0.956)
length seen up to 50 % of the longest run	—	0.851 (0.829–0.873)
Robometer-4B, success probability	0.818 (0.790–0.843)	0.821 (0.794–0.847)
Robometer-4B, last progress (descriptive)	0.838 (no CI)	—
Robometer-4B fine-tuned on ArmnetBench, success probability (in-sample)	0.994 (0.991–0.996)	0.994 (0.992–0.997)
TOPReward, success score	0.653 (0.623–0.684)	0.661 (0.631–0.692)
TOPReward, Yes/No component (descriptive)	—	0.678

Table 1: ArmnetBench, failure AUROC with 95 % episode-bootstrap CIs. Read down the within-policy column: no zero-shot reward-model score reaches length, and the fine-tune row is in-sample.

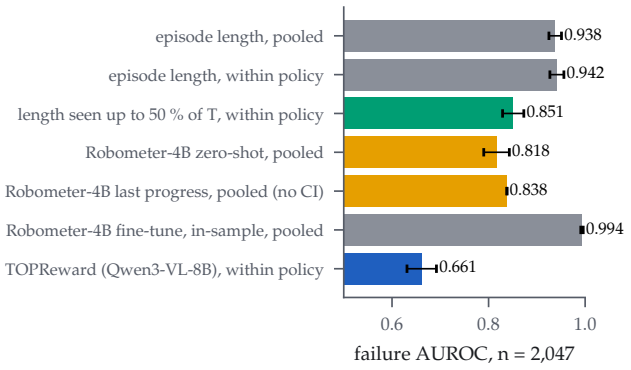


Figure 1: Failure AUROC on ArmnetBench: episode length against two zero-shot reward models. Length leads both, pooled and within policy (95 % CIs where available).

0.794–0.847). Its paired difference to length has a 95 % CI of -0.150 to -0.094. We ran TOPReward [1] with Qwen3-VL-8B [11] and no training. It scores 0.661 within policy. Its paired difference to length has a 95 % CI of -0.31 to -0.25. The protocols fixed each model's primary score before any score existed. For Robometer it is the success probability, and for TOPReward the success score of its paper. Two other scores from the same runs are in Table 1 as descriptive rows. They are higher, and still below length.

We also scored the released ArmnetBench fine-tune of Robometer, `jesbu1/robometer-4b-fft-armnet-tiled`, since the obvious reply to a zero-shot result is to use it. Its model card lists a dataset of all 2,499 ArmnetBench episodes, with no held-out split, as its training data. Every episode we score was therefore a training episode with its label, and the result is in-sample. On the 2,047 rollouts it scores 0.994 pooled and 0.994 within policy, 0.052 above length (paired CI 0.039 to 0.067). This shows that the labels can be fit. It does not show that a reward model beats length on rollouts it has not seen. We gave it

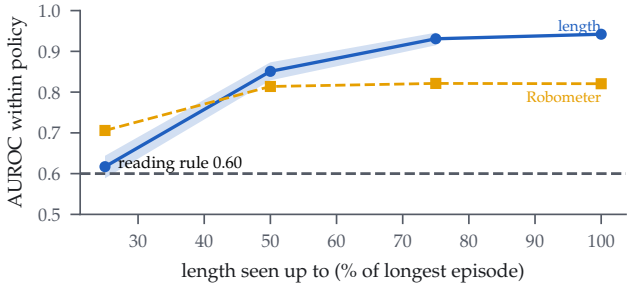


Figure 2: Length observed up to a fraction of the task's longest episode, AUROC within policy: most of the signal is there by half the episode (band: 95 % CI).

8 frames spread over the 64 of each tiled video, the length it was trained on. Scoring all 64 frames on 160 episodes gives 0.995 pooled against 0.989 for 8 frames.

4 Length is more than the timeout

Only 8.7 % of failures run as long as the task's longest episode, which we take as its timeout. At half that length, 88 % of failures are still running, and only 19 % of successes are. Length observed up to that cut gives 0.851 within policy (Fig. 2). The CI is 0.829–0.873. TOPReward on frames up to the same cut reaches 0.667. Robometer's success probability at the last frame before the cut gives 0.814 within policy (CI 0.787–0.839), which is below length at that cut. At a quarter of the longest episode the order is the other way, 0.706 for Robometer against 0.617 for length, and by three quarters length is ahead again, 0.931 against 0.821. These are prefixes of one pass over the whole episode (§7). Failed rollouts are slow, not timed out. Both readings fixed in the protocol hold.

On our own operator-stopped rollouts, length is at chance. It gives 0.56 on the 69 rollouts TOPReward scored and 0.52 on all 88. Here the policy-identity null is the baseline to beat. 17 of the 25 failures in the 69 come from one policy, TurboVLA [16]. Neither reward model beats policy identity. TOPReward scores 0.82 (CI 0.70–0.93), and policy identity alone 0.87. On the 50 episodes without TurboVLA, TOPReward scores 0.78. Robometer scores 0.577 on all 88, and policy identity 0.878.

5 Judges and monitors on our rollouts

Table 2 gives each judge and monitor a threshold-free AUROC where one is defined. It also gives failures caught and successes flagged at the operating point. For the judges at rest, that point is chosen in-sample.

We first ran a deterministic top-camera check (colour masks and geometry, no learning). It is right on 94.3 % of judged episodes. It has 0 false positives (Wilson 0–13 then gave Flora-Rewarder-27B [8] a 64-frame two-view

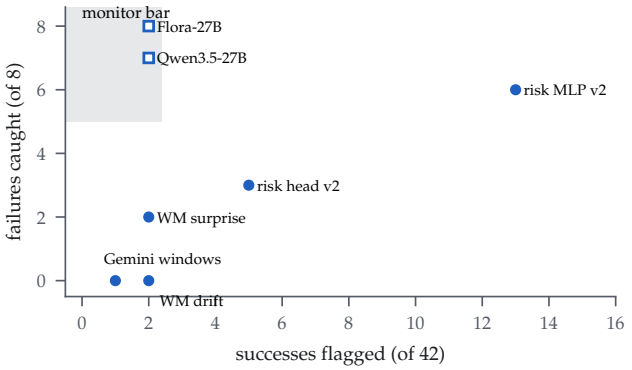


Figure 3: Failures caught against successes flagged on the 50 rollouts. No monitor (circles) reaches its shaded bar; the judges at rest (squares) sit at their in-sample operating point, where the monitor bar does not apply.

clip and NVIDIA's prompt. It reaches a threshold-free AUROC of 1.00 (bootstrap CI 1.00–1.00). At its in-sample operating point it flags 8 of 8 failures at 2 of 42 successes (Wilson on the false alarms 1–16 at least 6 of 8 failures at no more than 2 of 42 successes, so it passes. Its base model, Qwen3.5-27B [10], has no robot fine-tune. On the same input it also passes, with AUROC 0.91 (0.78–1.00) and 7 of 8 at 2 of 42. The input format mattered more than the fine-tune. Qwen3.5-9B [10] flags 16 successes and fails. Gemini 3.8 Flash [2] fails by one false alarm.

The bar for mid-episode monitors is at least 5 of 8 failures at no more than 1 of 20 false interventions. No monitor reaches it on these 50 rollouts (Fig. 3). The surprise of a separate world model, trained in the LeWorldModel recipe [7], catches 2 of 8 at 2 of 42 (AUROC 0.86). A Gemini sliding-window judge catches 0. We also built an action-conditioned risk head on the world model's latent. Its first version reached AUROC 0.93. It read elapsed time, so we withdrew that result. We pre-registered a second version without time or length. It catches 3 of 8 at 5 of 42. Its AUROC is 0.79, and 0.71 with shuffled actions. The MLP variant catches 6 of 8 and flags 13 of 42.

6 Related work

Robometer [5] scales a general-purpose reward model through trajectory comparisons. TOPReward [1] reads the token probabilities of a video-language model as hidden rewards. A survey of verifiers for robot policies [14] covers the wider family. FAIL-Detect [17] and SAFE [3] learn uncertainty- and feature-based failure scores for imitation and VLA policies. FARM [9] reads failure signals from the internal states of a frozen world model, like our world-model monitors. Our risk head follows ReAP [6], which predicts action-conditioned risk. VLA-Corrector [13] verifies stages, Spotter [4] adds a reflecting VLM, and REMORA [15] tracks progress through errors and recovery. ArmnetBench [12] and a standardized SO-101 VLA benchmark [18] evaluate policies on the same arm. We found no report of the elapsed-length baseline under the dataset's stopping rule in any of them.

7 Limitations

The judge pass on our own rollouts rests on 8 failures. Its operating thresholds were chosen on the same 50 episodes. A single operator, the author, labelled them and was not blind to the policy. The threshold-free AUROCs do not depend on the threshold. Fresh rollouts scored at the frozen thresholds would confirm the operating points. 17 of the 25 failures in our 69 rollouts are TurboVLA's. Any score that reads the policy therefore gets the policy-identity null for free. The risk head is trained and tested on the same 50 episodes. Five folds by episode keep episodes from leaking, but tasks and policies leak. The fine-tune comparison is in-sample, because the release has no held-out episodes, and its config needed three training-only keys removed to load in the pinned code. Robometer's truncated scores are prefixes of one pass over the whole episode. A frame's value may depend on later frames, so a pass over a clip that ends at the cut could give a different number. The 462 episodes that ran out of memory in the first pass were scored in a second pass on a larger GPU, with every frame kept. TOPReward's paper does not give its prefix count, frames per query or Yes/No wording, so those are our choices. The ArmnetBench result covers eight tasks. The timeout is approximated by each task's longest episode.

8 Conclusion

On ArmnetBench, where runs end at a time limit, episode length beat both zero-shot reward models, and a fine-tune beat it only on the episodes it was trained on. On our operator-stopped rollouts length was at chance, and a judge that saw the whole finished episode caught the failures. When you evaluate a reward model as a failure detector, report episode length within policy on the same rows and say how the episodes ended.

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method	kind	AUROC	caught / 8	flagged / 42	threshold
colour judge (no learning)	at rest	— (94.3 % right, 0 FP, 5 FN)	—	—	fixed rule
Flora-Rewarder-27B	at rest	1.00 (1.00–1.00)	8	2	in-sample
Qwen3.5-27B, same input	at rest	0.91 (0.78–1.00)	7	2	in-sample
Qwen3.5-9B, same input	at rest	0.75	7	16	in-sample
Gemini 3.8 Flash, same input	at rest	0.96	8	3	in-sample
Gemini sliding windows	monitor	—	0	1	fixed before scoring
world-model surprise	monitor	0.86	2	2	fixed before scoring
world-model latent drift	monitor	0.30	0	2	fixed before scoring
risk head, logistic (v2)	monitor	0.79	3	5	5-fold by episode
risk head, MLP (v2)	monitor	—	6	13	5-fold by episode

Table 2: Judges and monitors on the 50 operator-stopped rollouts (42 successes, 8 failures). Judges see the finished episode; monitors are held to $\geq 5/8$ caught at $\leq 1/20$ false interventions.

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analysis	repro folder	expected files	protocol
length baseline, truncation	repro/ length-null	length_ armnetbench.json, counts_ci.json	LENGTH_ PROTOCOL.md, frozen `1a69f30`
Robometer on ArmnetBench	repro/ robometer-judge, repro/ length-null	g9a_ab_ complete.json, robometer_ length.json	PROTOCOL.md (primary score fixed there); length-null amendment 2026-11-05
Robometer fine-tune on ArmnetBench (in-sample)	repro/ robometer-judge	ft_rows.jsonl, ft_ results.json	PROTOCOL_ FT.md (frozen before any score; deviations numbered)
Robometer and length on own rollouts	repro/ robometer-judge	g9a_own.json	PROTOCOL.md
colour judge	repro/ robometer-judge	g9c3_test_results.json	PROTOCOL.md
TOPReward, both datasets	repro/ topreward	topreward.json, own_rollout_ labels.csv	TOPREWARD_ PROTOCOL.md, `6241570`
judges at rest, input-format gate	repro/flora-input-format	g_flora_ summary.json, g_ flora_abl_ summary.json	PROTOCOL.md
window judge, world-model monitors	repro/ six-monitors	judge_tau.json, w1a_ summary.json	PROTOCOL.md
action-conditioned risk head	repro/reap	monitors.json, monitors_v2.json	PROTOCOL.md

Appendix: from plain name to analysis

Every number is recomputable from the repository; this table maps each analysis to its files.